

Assignment 7 - IoT visualization

17.3.2024

Sent 100%

Forsøk 1



Se gjennom tilbakemelding

2.4.2024

Forsøk 1 Poengsum:

100%



Legg til kommentar

Anonym vurdering: **nei**

Ubegrenset antall forsøk tillatt

16.12.2023 til 29.4.2024

▼ Detaljer

Preparation

This assignment assumes you have completed the Wi-Fi assignment. The resources and general instructions from previous assignments are still useful here, this assignment will only list new relevant resources.

1. Follow the instructions from [Assignment 6 - Wi-Fi \(https://uia.instructure.com/courses/14984/assignments/58909\)](https://uia.instructure.com/courses/14984/assignments/58909) to create a new Mbed OS program under your git folder (`<course folder>/assignments/`) called `assignment_7`. Use same Wi-Fi network configuration as in previous assignment.
2. Edit `mbed_app.json` and increase the stack size by 2x or 4x. This can help prevent stack overflow errors.
3. If you don't have Docker on your current machine, you must install Docker Desktop from [this website](https://docs.docker.com/desktop/)  (<https://docs.docker.com/desktop/>). Select target OS, check system requirements found on the website and follow the install instructions.

Note: For Windows machines you should run the Docker Desktop Installer program as **Administrator** and make sure "Use WSL 2 instead of Hyper-V (recommended)" is checked.

Resources

- See the lecture for relevant resources and examples.
- [Wi-Fi error codes](https://os.mbed.com/docs/mbed-os/v6.16/mbed-os-api-doxy/nsapi__types__8h_source.html)  (https://os.mbed.com/docs/mbed-os/v6.16/mbed-os-api-doxy/nsapi__types__8h_source.html)

Requirements

The group should start by reading temperature and humidity values from the HTS221 sensor on the board as described in [Assignment 3 - I2C: Temperature and humidity \(https://uia.instructure.com/courses/14984/assignments/58906\)](https://uia.instructure.com/courses/14984/assignments/58906). Test by printing the values so you can

see this part of the program is working.

Once you are able to read the sensor values, you will install ThingsBoard and report the sensor values to ThingsBoard using Wi-Fi. ThingsBoard will then visualize these values.

1. 10% Install ThingsBoard on your PC using Docker (see lecture)
2. 15% Log in to ThingsBoard as a tenant (see lecture) and add a device
3. 15% Make a ThingsBoard dashboard with two widgets (one showing temperature, the other humidity)
4. 30% Report temperature to ThingsBoard and view value on the dashboard
5. 30% Report humidity to ThingsBoard and view value on the dashboard

Optional: Also read and report the barometer value. This exercise is left up to the reader.

Delivery

Deliverer the following as text in Canvas:

- Link to Bitbucket where to code has been delivered.
Directly to the file if there is only 1, or the directory if more than 1.
- Link to YouTube (or other video service) with video showing each of the requirements.
Make sure that each requirement you have completed is clearly shown.
You don't need to show the actual code, just which of the requirements are met.